

Practical No. 6: Travelling Salesman Problem using Greedy Approach in C

Program in C

```
#include <stdio.h>
#define N 4
#define INF 9999

int graph[N][N] = {
    {0, 10, 15, 20},
    {10, 0, 35, 25},
    {15, 35, 0, 30},
    {20, 25, 30, 0}
};

int visited[N];

int findNearestCity(int currentCity)
{
    int nearest = -1;
    int minDist = INF;
    int city;

    for (city = 0; city < N; city++)
    {
        if (!visited[city] && graph[currentCity][city] < minDist)
        {
            minDist = graph[currentCity][city];
            nearest = city;
        }
    }
    return nearest;
}

void tspGreedy(int start)
{
    int currentCity = start;
    int totalCost = 0;
    int nextCity;
    int i;

    for (i = 0; i < N; i++)
        visited[i] = 0;

    visited[currentCity] = 1;
    printf("Path: %d", currentCity);
```

```

    for (i = 0; i < N - 1; i++)
    {
        nextCity = findNearestCity(currentCity);

        visited[nextCity] = 1;
        totalCost += graph[currentCity][nextCity];

        printf(" -> %d", nextCity);
        currentCity = nextCity;
    }

    totalCost += graph[currentCity][start];
    printf(" -> %d\n", start);

    printf("Total cost using greedy approach = %d\n", totalCost);
}

int main()
{
    tspGreedy(0);
    return 0;
}

```

Sample Output

Path: 0 -> 1 -> 3 -> 2 -> 0
 Total cost using greedy approach = 80